In [3]:
```
text = """Text analytics is the process of analyzing textual data to e
It involves preprocessing steps such as tokenization, stopword removal

print(text)
```

```
Text analytics is the process of analyzing textual data to extract me
aningful insights.
It involves preprocessing steps such as tokenization, stopword remova
l, stemming, and lemmatization.
```

In [4]:
```
tokens = word_tokenize(text)
print(tokens)
```

```
['Text', 'analytics', 'is', 'the', 'process', 'of', 'analyzing', 'tex
tual', 'data', 'to', 'extract', 'meaningful', 'insights', '.', 'It',
'involves', 'preprocessing', 'steps', 'such', 'as', 'tokenization',
',', 'stopword', 'removal', ',', 'stemming', ',', 'and', 'lemmatizati
on', '.']
```

In [5]:
```
pos_tags = pos_tag(tokens)
print(pos_tags)
```

```
[('Text', 'NN'), ('analytics', 'NNS'), ('is', 'VBZ'), ('the', 'DT'),
('process', 'NN'), ('of', 'IN'), ('analyzing', 'VBG'), ('textual', 'J
J'), ('data', 'NNS'), ('to', 'TO'), ('extract', 'VB'), ('meaningful',
'JJ'), ('insights', 'NNS'), ('.', '.'), ('It', 'PRP'), ('involves',
'VBZ'), ('preprocessing', 'VBG'), ('steps', 'NNS'), ('such', 'JJ'),
('as', 'IN'), ('tokenization', 'NN'), (',', ','), ('stopword', 'NN'),
('removal', 'NN'), (',', ','), ('stemming', 'VBG'), (',', ','), ('an
d', 'CC'), ('lemmatization', 'NN'), ('.', '.')]
```

In [6]:
```
stop_words = set(stopwords.words('english'))

filtered_tokens = [word for word in tokens if word.lower() not in stop

print(filtered_tokens)
```

```
['Text', 'analytics', 'process', 'analyzing', 'textual', 'data', 'ext
ract', 'meaningful', 'insights', '.', 'involves', 'preprocessing', 's
teps', 'tokenization', ',', 'stopword', 'removal', ',', 'stemming',
',', 'lemmatization', '.']
```

In [7]:
```
stemmer = PorterStemmer()

stemmed_words = [stemmer.stem(word) for word in filtered_tokens]

print(stemmed_words)
```

```
['text', 'analyt', 'process', 'analyz', 'textual', 'data', 'extract',
'meaning', 'insight', '.', 'involv', 'preprocess', 'step', 'token',
',', 'stopword', 'remov', ',', 'stem', ',', 'lemmat', '.']
```

In [8]:
```
lemmatizer = WordNetLemmatizer()

lemmatized_words = [lemmatizer.lemmatize(word) for word in filtered_to

print(lemmatized_words)
```

```
['Text', 'analytics', 'process', 'analyzing', 'textual', 'data', 'ext
ract', 'meaningful', 'insight', '.', 'involves', 'preprocessing', 'st
ep', 'tokenization', ',', 'stopword', 'removal', ',', 'stemming',
',', 'lemmatization', '.']
```

In [9]:
```
documents = [
    text,
    "Text analytics helps in extracting useful insights from text data
    "Preprocessing includes tokenization and cleaning of text."
]
```

In [10]:
```
vectorizer = TfidfVectorizer()

tfidf_matrix = vectorizer.fit_transform(documents)

feature_names = vectorizer.get_feature_names_out()
```

In [11]:
```
tfidf_df = pd.DataFrame(tfidf_matrix.toarray(), columns=feature_names)

tfidf_df
```

Out[11]:

	analytics	analyzing	and	as	cleaning	data	extract	extracting	from
0	0.164411	0.216181	0.164411	0.216181	0.000000	0.164411	0.216181	0.000000	0.000000
1	0.266720	0.000000	0.000000	0.000000	0.000000	0.266720	0.000000	0.350704	0.350704
2	0.000000	0.000000	0.352215	0.000000	0.463121	0.000000	0.000000	0.000000	0.000000

3 rows × 32 columns

In [ ]: